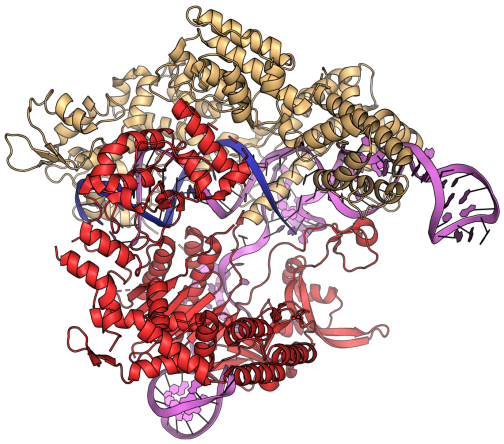
DNA Mutation Analysis



As part of a new cost-cutting initiative, UVA Health decided to replace all of their doctors and nurses with AI agents. But oh no! All data centers in Virginia were shut down by the governor after protests in Northern Virginia, and in retaliation, Google, Anthropic, and OpenAI have blocked the state from using their AI agents. Instead of bringing back human medical staff, UVA Health doubled down and hired a team of software engineers to simply build an in-house LLM from scratch. In this job market, you decided to take the position, even though you have absolutely no machine learning experience.

On your first day, your boss panics because the system has no way of diagnosing genetic disorders. He tells you that the lab can sequence a patient's genome into a text file, and the AI agent just needs to figure out the genetic disorder from that. He gives you a day to complete this task and assigns it 3 story points (he has no technical experience). Hopefully your new coworkers know what they're doing but rats! It turns out their qualifications are being related to someone on the board of visitors.

You think you remember from BIO 2100 these disorders are caused by a mutation in a DNA sequence. You track down one of the recently laid-off nurses to ask if there are specific sequence patterns that flag these disorders. They confirm you are right and hand you a list of known pathogenic genetic markers. Additionally, they tell you about a new automated CRISPR gene-editing rig in the basement that can automatically splice and fix mutated DNA. The machine only needs the exact target sequence (the mutated pattern) and its starting index in the genome so it knows where to execute the replacement.

Armed with this information, rather than confessing to your boss that you can't train a neural network, you decide to just construct a multi-pattern string matching algorithm. Given a patient's DNA sequence as a string, and a list of mutation patterns, you need to return the exact index of each mutation in the DNA sequence alongside the pattern itself.

Input

The first line of input will be a string representing a patient's sequenced genome. This string will have a maximum length of 10^5, with an alphabet of nucleotide bases {A, T, C, G}. The second line of input will be a number $1 \leq M \leq 1000$, the number of target mutations you must check for. The subsequent M lines each describe a mutation. Each line contains the name of the mutation and its corresponding sequence pattern, separated by a single space.Each mutation genome pattern will be strings of length $1 \leq S \leq 32$, with an alphabet of nucleotide bases {A, T, C, G}.

Output

Please output the target mutations that are present in the patient's genome. Each detected mutation must be printed on a new line, containing the mutation's name followed by a single space and the starting index (0-indexed) of its first nucleotide base in the genome.

Format: <Mutation Name> <Starting Index>

Sample Input	Sample Output
CCGTGCACCTAGATACAGATCAGCAGCAGT 4 Sickle Cell Trait GTGCACCT Huntingtons Marker CAGCAGCAG Breast Cancer Risk GATACAGAT Cystic Fibrosis Gene AAATTTCCC	Sickle Cell Trait 2 Breast Cancer Risk 11 Huntingtons Marker 20